

The Impact of the Russia-Ukraine War on Aviation Activity: Evidence from a Difference-in-Differences Approach

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This paper estimates the impact of the Russia-Ukraine airspace closure on European aviation activity using route-month panel data from Eurostat and a difference-in-differences design. We find that affected routes experienced significant declines in flights, passengers, and seat capacity. Estimated CO₂ emissions declined less than flight volumes, suggesting that rerouting partially offset emissions reductions.

Russian invasion of Ukraine on 24 February 2022 triggered a rapid sequence of mutual airspace bans, effectively closing the shortest corridor between Europe and Asia. Existing research shows that the closure substantially increased rerouting distances and fuel consumption for European carriers (Ostroumov et al., 2025; Dannet, Bellouin, and Boucher, 2025). This paper examines the causal effect of the airspace closure on Europe-to-Asia flight operations using route-month panel data from Eurostat and a difference-in-differences framework. We estimate the impact on flight frequency, passenger volumes, and estimated CO₂ emissions.

I. Literature Review

Existing studies document the operational consequences of the closure, including increased rerouting distances, airline costs, and emissions (Ostroumov et al., 2025; Ivannikova, Sokolova, and Cherednichenko, 2024; Dannet, Bellouin, and Boucher, 2025). Ennen and Wozny (2024) provide the closest related study, applying a DID framework to show that Europe-Asia airfares increased following the closure. However, existing work has not examined the causal effect on route-level flight activity, passenger volumes, or estimated emissions. Our empirical strategy follows established DID approaches used to evaluate shocks in aviation markets (Bernardo and Fageda, 2017; Ito and Lee, 2005).

II. Data

We construct a monthly panel of flight routes using the European Commission’s Aviation Passenger Transport dataset. The panel covers 31 European countries from January 2015 through December 2024, comprising 370 airports and 9,783 unique origin-destination pairs. The sample spans Northern Europe, Central Europe, Southern Europe, and the Balkans.

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We analyse data at the route-month level, where each observation represents a unique origin-destination pair in a particular calendar month. Our final panel contains 783,057 route-month observations (excluding COVID years). For each route-month, we observe: (i) number of flights operated, (ii) total passengers carried, (iii) available seats, and (iv) load factor (passenger-to-seat ratio).

Routes are classified as treated if the great-circle path transits either the Russian or Ukrainian Flight Information Region (FIR), using ICAO FIR boundary polygons – the legally defined units within which the airspace closures were enacted¹. Treatment is binary and deterministic based on geography. Figure 1 shows the geographic distribution.

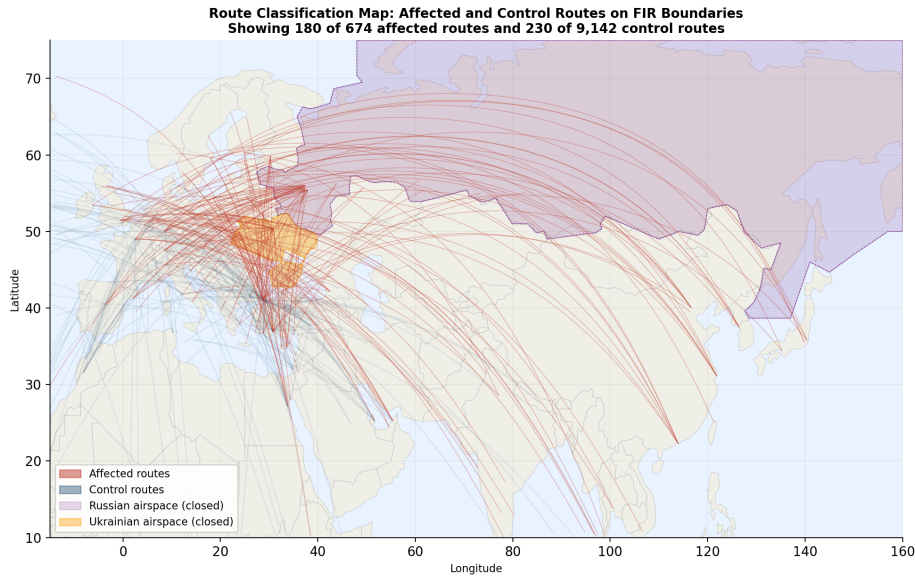


FIGURE 1. ROUTE CLASSIFICATION: TREATED VS. CONTROL ROUTES

Note: Red lines indicate routes transiting Russian and Ukrainian Flight Information Regions, shaded in purple and orange. Blue lines indicate control routes avoiding closed airspace. The sample includes 674 treated routes and 9,109 control routes.

COVID-period observations are excluded as pandemic disruptions were route-heterogeneous and correlated with treatment, threatening parallel trends in ways a time dummy cannot address; Panel B of Table 4 confirms results are robust to their inclusion. UK routes are dropped post-December 2019 due to Brexit-related changes in reporting.

We compute log transformations of flights, passengers, and seats for elastic-

¹Great-circle paths are approximated using the Haversine formula, interpolated at 100 equally spaced points. FIR boundaries are defined using ICAO polygon; a route is treated if any interpolated point falls within either FIR polygon.

ity interpretation. Load factor indicates whether airlines adjusted capacity proportionally to demand. CO₂ emissions are estimated from flight counts using distance-band fuel burn averages following ICAO’s route-level methodology, converted to CO₂ using a factor of 3.16, and adjusted for rerouting using exposure-group multipliers; Table 5 confirms robustness to alternative multiplier assumptions².

Table 1 confirms treated routes are structurally different – longer and served by larger aircraft – making route fixed effects necessary for identification. Importantly, pre-closure differences in flight volumes and passengers are stable, supporting the parallel trends assumption. Post-closure declines in flights and passengers on treated routes, with load factors unchanged, suggest airlines adjusted capacity proportionally to demand.

TABLE 1—SUMMARY STATISTICS BY TREATMENT STATUS AND PERIOD

Variable	Pre-Closure (2015–2019)			Post-Closure (Feb 2022–Dec 2024)		
	Control	Treated	Diff	Control	Treated	Diff
<i>Flight Operations</i>						
Flights per route-month	134.0	98.4	−35.5***	115.8	82.7	−33.0***
Passengers per route-month	16,553	18,419	+1,866***	16,352	17,350	+998***
Seats per route-month	21,639	23,972	+2,333***	20,139	21,474	+1,334***
Load factor	0.808	0.795	−0.014***	0.819	0.821	+0.003
<i>Route Characteristics</i>						
Distance (km)	1,679	5,076	+3,397***	1,720	5,613	+3,893***

Note: Treated routes transit Russian or Ukrainian Flight Information Regions. Control routes avoid closed airspace. Differences are tested using two-sample t-tests. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

III. Methodology

We use the Two-Way Fixed Effects (TWFE) difference-in-differences (DiD) estimator to exploit the geographic discontinuity created by the closure. We estimate the TWFE model using ordinary least squares with route and month-year fixed effects:

$$Y_{rt} = \alpha_r + \gamma_t + \beta \cdot (\text{Treated}_r \times \text{Post}_t) + \varepsilon_{rt}$$

where: Y_{rt} is the outcome for route r in month-year t , α_r and γ_t are route and time fixed effects respectively³ that would otherwise confound the DiD estimate. Both

²CO₂ emissions are estimated by multiplying flight counts by distance-band fuel burn averages (kg per flight), converted to CO₂ using a factor of 3.16 kg CO₂ per kg fuel. Distance bands are defined as short (less than 1,000 km), medium (1,000–3,000km), long (3,000–7,000km), and very long (greater than 7,000 km). Rerouting multipliers of 1.01, 1.16, and 1.43 are applied to low, medium, and high-exposure routes respectively to account for additional fuel burn on lengthened post-closure flight paths.

³Route fixed effects absorb time-invariant route-specific heterogeneity. Time fixed effects capture aggregate trends common to all routes.

are estimated at the canonical (undirected) route-pair level. Treated_r indicates route transiting closed airspace, Post_t indicates months after February 2022, and β is the average treatment effect on the treated (ATT). Standard errors are clustered at the route-pair level, as residuals are likely correlated within a route across time but independent across routes.

Our identification strategy relies on the assumption that treated and control routes would have followed parallel trends absent treatment. Figure 2 validates this - Pre-treatment coefficients fluctuate around zero with no clear systematic pre-trend or anticipatory effects. The sharp break occurs in February 2022 aligning with the closure date. Table 4 reports a placebo test using a false treatment date (February 2019). Log passengers and the load factor are both near-zero and insignificant. Log flights and seats show significant placebo effects, though their magnitudes are considerably smaller than post-closure estimates, providing no evidence of systematic pre-treatment variation.

Treatment assignment is determined entirely by geography and is orthogonal to airline characteristics. The closure was sudden and unexpected, minimising anticipatory behaviour. Furthermore, we assume treatment of one route does not affect control routes. While some general equilibrium effects are plausible, for example airlines reassigning aircraft from disrupted to unaffected routes, any such effects would bias our estimates toward zero, making our results a lower bound on the true treatment effect.

COVID-period observations from 2020–2021 are excluded because pandemic-related disruptions were highly heterogeneous across routes and potentially correlated with treatment status. We additionally assess robustness to alternative sample definitions and rerouting multiplier assumptions for estimated CO₂ emissions.

TABLE 2—EFFECT OF AIRSPACE CLOSURE ON ROUTE-LEVEL OUTCOMES

Outcome	Coefficient	Approx. Effect	Observations
Log Flights	-0.0768*** (0.0295)	-7.40%	779,438
Log Passengers	-0.1642*** (0.0309)	-15.14%	734,912
Log Seats	-0.1040*** (0.0292)	-9.88%	760,871
Load Factor	-0.0050 (0.0053)	-0.0050	713,424
Log CO ₂	-0.0269 (0.0276)	-2.65%	768,217

Note: Route-level TWFE estimates. Treatment equals $\text{Treated} \times \text{Post}$. Standard errors clustered at the route-pair level are reported in parentheses. All specifications include route and month-year fixed effects. Route pairs range from 9,309 to 9,752 across specifications. For logged outcomes, approximate percentage effects are calculated as $100 \times (\exp(\beta) - 1)$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IV. Results

A. Main Specification

Table 2 presents the baseline two-way fixed effects estimates. The estimates indicate a statistically significant decline in flight activity on affected routes. Log flights declined by 0.077 log points, equivalent to an approximate 7.4% reduction relative to unaffected routes. The war appears to have the greatest negative impact on passengers, with an approximate reduction of 15.1%. Seat capacity also declined significantly but there was no significant impact on load factor, suggesting that airlines adjusted capacity in line with passenger demand.

Our estimates suggest a reduction of approximately 2.7% in estimated log CO₂ emissions at the route level compared to unaffected routes, although this estimate is not statistically significant at conventional levels. However, the smaller magnitude relative to the reduction in flights suggests that rerouting effects may have partially offset emissions reductions through longer flight paths. These estimates should be considered a lower-bound as this does not capture indirect emissions from passengers rerouting via alternative connecting hubs.

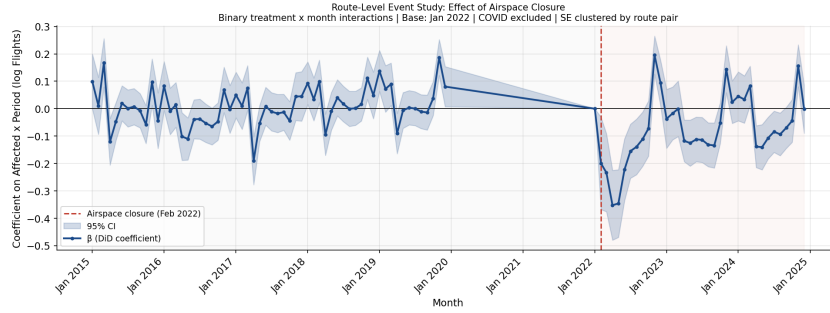


FIGURE 2. EVENT STUDY: EFFECT OF AIRSPACE CLOSURE ON FLIGHTS

Note: The figure reports monthly event-study coefficients from the route-level TWFE specification. January 2022 is the omitted base period. Error bars represent 95% confidence intervals based on route-clustered standard errors.

B. Event Study

Figure 2 validates the parallel trends assumption - pre-treatment coefficients are small, stable and insignificant (max coefficient = 0.1898), showing no clear evidence of systematic pre-trends or anticipatory effects. The sharp break occurs in February 2022 aligning with the closure date. Table 4, reports a placebo test using a false treatment date (February 2019), yielding near-zero, insignificant coefficients (e.g., log passengers $\beta = +0.0164$, $p = 0.254$), confirming the effect is specific to the actual closure.

TABLE 3—HETEROGENEITY ANALYSIS

	Log Flights	Log Passengers	Log Seats	Load Factor	Log CO ₂
<i>Panel A. Exposure Heterogeneity</i>					
Russia Exposure	-0.3028*** (0.0426)	-0.3797*** (0.0461)	-0.3363*** (0.0416)	-0.0303*** (0.0098)	-0.1794*** (0.0403)
Ukraine Exposure	0.0386 (0.0367)	0.0005 (0.0380)	0.0151 (0.0377)	0.0108* (0.0061)	0.0444 (0.0364)
Both Exposure	0.0276 (0.0870)	-0.3049*** (0.0731)	0.0019 (0.0762)	-0.0024 (0.0137)	0.0663 (0.0837)
<i>Panel B. Distance-Band Heterogeneity</i>					
Medium (< 3000 km)	0.1970*** (0.0503)	0.1396** (0.0598)	0.1970*** (0.0507)	-0.0040 (0.0092)	0.1976*** (0.0502)
Long (3000–7000 km)	-0.0876* (0.0519)	-0.0947** (0.0449)	-0.1222*** (0.0442)	0.0198** (0.0094)	-0.0758 (0.0516)
Very Long (> 7000 km)	-0.2551*** (0.0374)	-0.3443*** (0.0392)	-0.2965*** (0.0356)	-0.0188** (0.0077)	-0.1508*** (0.0347)

Note: All specifications include route and month-year fixed effects. Standard errors clustered at the route-pair level are reported in parentheses. COVID-period observations are excluded. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C. Heterogeneity and Mechanism Analysis

From Table 3 the disruption appears primarily driven by exposure to Russian airspace rather than Ukrainian airspace. Panel A shows that routes exposed to Russian airspace experienced substantially larger declines across flights, passengers, seats, and estimated CO₂ emissions, while Ukraine-only exposure produced generally small and insignificant effects.

Panel B shows very long-haul routes experienced the largest declines across aviation outcomes, consistent with greater reliance on Russian overflight corridors prior to the invasion. We observe a statistically significant increase in flights, passengers, seats, and estimated CO₂ emissions for routes with a pre-war distance of around 3,000 km. One possible explanation is a substitution effect in which airlines and passengers shifted demand away from disrupted long-haul routes to relatively more reliable and lower-cost medium-haul journeys within Europe and parts of West Asia.

D. Robustness Checks

Table 4 reports several robustness checks. The main results remain qualitatively similar when COVID-period observations are included, although coefficient magnitudes change somewhat due to pandemic-related disruptions to aviation demand. Third, excluding Finland – one of the European countries most exposed to Russian overflight restrictions – does not materially alter the main findings, indicating that the results are not driven solely by Finnish routes. Finally, alternative rerouting multiplier assumptions for estimated CO₂ emissions produce

TABLE 4—ROBUSTNESS CHECKS

Outcome	Coefficient	Standard Error	p-value
<i>Panel A. Placebo Treatment Date: February 2019</i>			
Log Flights	0.0296**	(0.0131)	0.0239
Log Passengers	0.0164	(0.0144)	0.2542
Log Seats	0.0437***	(0.0145)	0.0026
Load Factor	-0.0016	(0.0031)	0.5985
Log CO ₂ (adj. est.)	0.0288**	(0.0131)	0.0282
<i>Panel B. Full Sample Including COVID Years</i>			
Log Flights	-0.0826***	(0.0267)	0.0020
Log Passengers	-0.1291***	(0.0284)	0.0000
Log Seats	-0.1322***	(0.0268)	0.0000
Load Factor	0.0080	(0.0055)	0.1486
Log CO ₂ (adj. est.)	-0.0351	(0.0251)	0.1613
<i>Panel C. Excluding Finland</i>			
Log Flights	-0.0644**	(0.0300)	0.0320
Log Passengers	-0.1497***	(0.0314)	0.0000
Log Seats	-0.0950***	(0.0300)	0.0015
Load Factor	-0.0002	(0.0052)	0.9752
Log CO ₂ (adj. est.)	-0.0204	(0.0284)	0.4730

Note: Panel A reports estimates using a placebo treatment date of February 2019 and the pre-2022 sample. Panel B reports estimates using the full sample including COVID years. Panel C reports estimates excluding Finland to assess whether results are driven by Finland's disproportionate exposure to Russian airspace. All specifications include route and month-year fixed effects. Standard errors are clustered by route. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

similar qualitative conclusions, although the emissions estimates remain statistically imprecise across specifications.

V. Conclusion

This paper uses a difference-in-differences framework to estimate the causal impact of the closure of Russian and Ukrainian airspace on European aviation activity. Affected routes experienced significant declines in flights, passengers, and seat capacity following the closure. The absence of a significant change in load factors suggests that airlines adjusted capacity proportionately to passenger demand. The point estimate suggests emissions declined by less than flight volumes, consistent with rerouting over longer alternative corridors partially offsetting emissions reductions from fewer flights, though this estimate is statistically imprecise and should be interpreted cautiously.

The largest effects occurred on routes exposed to Russian airspace and on very long-haul routes, reflecting their greater reliance on pre-war overflight corridors. The analysis has several limitations. Treatment is inferred from route geography rather than observed rerouted flight paths, CO₂ estimates rely on estimated rerouting multipliers, and the placebo test yields small but statistically significant coefficients for flights, seats, and estimated CO₂, suggesting some residual pre-trend variation, though magnitudes are substantially smaller than post-closure

TABLE 5—CO₂ MULTIPLIER SENSITIVITY

Scenario	Coefficient	Standard Error	<i>p</i> -value
Base	-0.0269	(0.0276)	0.3307
Low (-20%)	-0.0351	(0.0279)	0.2086
High (+20%)	-0.0190	(0.0274)	0.4896
Flat (all 1.10)	-0.0142	(0.0282)	0.6147

Note: This table reports the sensitivity of the estimated log CO₂ effect to alternative rerouting multiplier assumptions. All specifications include route and month-year fixed effects. Standard errors are clustered by route. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

estimates. Nonetheless, the results highlight how geopolitical shocks can generate lasting disruptions to international transport networks.

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Data Appendix

Dataset	Source	Variables used	Time period
Aviation Passenger Transport (avia_paocc)	Eurostat / European Commission	Flights operated, passengers carried, seats available, load factor	Jan 2015 – Dec 2024
ICAO Flight Information Region (FIR) Boundary Polygons	International Civil Aviation Organisation (ICAO)	FIR polygon boundaries for Russian and Ukrainian airspace; used to classify treated routes	At closure (Feb 2022)
Distance-band fuel burn averages	ICAO Carbon Emissions Calculator	Fuel burn per flight by distance band; CO ₂ conversion factor (3.16 kg CO ₂ /kg fuel)	N/A
OurAirports database	airport (ourairports.com)	Airport ICAO codes, latitude, longitude; used to compute great-circle route distances and FIR transit classification	Accessed 2025
Commercial flights by aircraft type (avia_tf_aca)	Eurostat / European Commission	Aircraft-type flight counts by airport-month; used to estimate route-level fuel burn and CO ₂	Jan 2015 – Dec 2024

Notes: All data are publicly available and require no registration to access. The Eurostat passenger dataset (`avia_par`) is available at ec.europa.eu/eurostat. The Eurostat aircraft-type dataset (`avia_tf_aca`) is available at the same portal. ICAO FIR boundaries are available at gis.icao.int/icaofir. ICAO fuel burn methodology is documented at [icao.int/Methodology/ICAO/Carbon/Emissions.pdf](https://www.icao.int/Methodology/ICAO/Carbon/Emissions.pdf). OurAirports coordinate data are available at ourairports.com. UK routes are excluded post-December 2019 due to Brexit-related changes in Eurostat reporting coverage. COVID-period observations (2020–2021) are excluded from the main sample. Great-circle paths are approximated using the Haversine formula interpolated at 100 equally spaced points; a route is classified as treated if any point falls within the Russian or Ukrainian FIR polygon.

APPENDIX

TABLE A1—FULL EVENT STUDY COEFFICIENTS

Period	Coefficient	SE	Period	Coefficient	SE	Period	Coefficient	SE
Jan 2015	0.0992*	(0.0511)	Sep 2017	-0.0128	(0.0335)	May 2022	-0.3459***	(0.0630)
Feb 2015	0.0104	(0.0531)	Oct 2017	-0.0444	(0.0358)	Jun 2022	-0.2226***	(0.0576)
Mar 2015	0.1678***	(0.0449)	Nov 2017	0.0444	(0.0445)	Jul 2022	-0.1550***	(0.0503)
Apr 2015	-0.1205***	(0.0434)	Dec 2017	0.0452	(0.0460)	Aug 2022	-0.1392***	(0.0517)
May 2015	-0.0469	(0.0352)	Jan 2018	0.0933**	(0.0396)	Sep 2022	-0.1114**	(0.0505)
Jun 2015	0.0198	(0.0327)	Feb 2018	0.0341	(0.0399)	Oct 2022	-0.0721	(0.0513)
Jul 2015	0.0013	(0.0345)	Mar 2018	0.1000**	(0.0396)	Nov 2022	0.1955***	(0.0355)
Aug 2015	0.0076	(0.0338)	Apr 2018	-0.0950**	(0.0385)	Dec 2022	0.0952**	(0.0421)
Sep 2015	-0.0072	(0.0318)	May 2018	-0.0096	(0.0353)	Jan 2023	-0.0368	(0.0549)
Oct 2015	-0.0580*	(0.0353)	Jun 2018	0.0398	(0.0337)	Feb 2023	-0.0170	(0.0491)
Nov 2015	0.0972**	(0.0431)	Jul 2018	0.0176	(0.0330)	Mar 2023	-0.0002	(0.0516)
Dec 2015	-0.0436	(0.0518)	Aug 2018	-0.0003	(0.0327)	Apr 2023	-0.1172***	(0.0388)
Jan 2016	0.0828*	(0.0456)	Sep 2018	0.0016	(0.0317)	May 2023	-0.1254***	(0.0429)
Feb 2016	-0.0095	(0.0433)	Oct 2018	0.0159	(0.0308)	Jun 2023	-0.1118***	(0.0416)
Mar 2016	0.0143	(0.0416)	Nov 2018	0.1112***	(0.0394)	Jul 2023	-0.1138***	(0.0419)
Apr 2016	-0.1014**	(0.0443)	Dec 2018	0.0487	(0.0435)	Aug 2023	-0.1314***	(0.0408)
May 2016	-0.1102***	(0.0392)	Jan 2019	0.1380***	(0.0382)	Sep 2023	-0.1340***	(0.0419)
Jun 2016	-0.0387	(0.0363)	Feb 2019	0.0725*	(0.0433)	Oct 2023	-0.0523	(0.0378)
Jul 2016	-0.0378	(0.0372)	Mar 2019	0.0895**	(0.0394)	Nov 2023	0.1441***	(0.0437)
Aug 2016	-0.0532	(0.0357)	Apr 2019	-0.0905**	(0.0376)	Dec 2023	0.0240	(0.0395)
Sep 2016	-0.0647*	(0.0342)	May 2019	-0.0062	(0.0333)	Jan 2024	0.0451	(0.0455)
Oct 2016	-0.0468	(0.0348)	Jun 2019	0.0037	(0.0320)	Feb 2024	0.0338	(0.0440)
Nov 2016	0.0687	(0.0423)	Jul 2019	0.0005	(0.0315)	Mar 2024	0.0839**	(0.0364)
Dec 2016	-0.0024	(0.0445)	Aug 2019	-0.0106	(0.0315)	Apr 2024	-0.1383***	(0.0389)
Jan 2017	0.0500	(0.0425)	Sep 2019	-0.0138	(0.0325)	May 2024	-0.1404***	(0.0421)
Feb 2017	0.0096	(0.0434)	Oct 2019	0.0369	(0.0317)	Jun 2024	-0.1069***	(0.0388)
Mar 2017	0.0758*	(0.0416)	Nov 2019	0.1868***	(0.0335)	Jul 2024	-0.0840**	(0.0384)
Apr 2017	-0.1898***	(0.0446)	Dec 2019	0.0806**	(0.0373)	Aug 2024	-0.0935**	(0.0397)
May 2017	-0.0539	(0.0334)	Jan 2022	0.0000	(base)	Sep 2024	-0.0698*	(0.0382)
Jun 2017	0.0092	(0.0352)	Feb 2022	-0.1990**	(0.0860)	Oct 2024	-0.0445	(0.0398)
Jul 2017	-0.0110	(0.0355)	Mar 2022	-0.2335***	(0.0726)	Nov 2024	0.1568***	(0.0391)
Aug 2017	-0.0173	(0.0355)	Apr 2022	-0.3522***	(0.0650)	Dec 2024	-0.0014	(0.0448)

Note: This appendix table reports the full set of monthly coefficients from the route-level event study specification. The dependent variable is log flights. Coefficients correspond to interactions between affected routes and each month-year period. January 2022 is the omitted base period. All specifications include route and month-year fixed effects. Standard errors are clustered by route. COVID-period observations are excluded. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.